COMP7024 Programming for Data Science

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```r
orders <- read_csv("project_orders.csv")
products <- read_csv("project_product_info.csv")

head(orders)
```

```
## # A tibble: 6 x 7
##   InvoiceNo StockCode Quantity InvoiceDate    UnitPrice CustomerID Country
##   <chr>     <chr>        <dbl> <chr>              <dbl>      <dbl> <chr>
## 1 536365    85123A           6 1/12/2010 8:26      2.55      17850 United Kingd~
## 2 536365    71053            6 1/12/2010 8:26      3.39      17850 United Kingd~
## 3 536365    84406B           8 1/12/2010 8:26      2.75      17850 United Kingd~
## 4 536365    84029G           6 1/12/2010 8:26      3.39      17850 United Kingd~
## 5 536365    84029E           6 1/12/2010 8:26      3.39      17850 United Kingd~
## 6 536365    22752            2 1/12/2010 8:26      7.65      17850 United Kingd~
```

```r
head(products)
```

```
## # A tibble: 6 x 2
##   StockCode Description
##   <chr>     <chr>
## 1 85123A    WHITE HANGING HEART T-LIGHT HOLDER
## 2 71053     WHITE METAL LANTERN
## 3 84406B    CREAM CUPID HEARTS COAT HANGER
## 4 84029G    KNITTED UNION FLAG HOT WATER BOTTLE
## 5 84029E    RED WOOLLY HOTTIE WHITE HEART.
## 6 22752     SET 7 BABUSHKA NESTING BOXES
```

#template to answer the questions

Q1

(a)

Explanation

The objective of the analysis is to evaluate the performance of the product by analyzing the total revenue generated. This is achieved by using the orders dataset, which has details of the quantity of each item purchased and the prices of each item. Additionally, the product information dataset has descriptions of each product. This data is combined by matching the StockCode column, allowing the descriptions to be added to the orders data.

Before performing the analysis, cleaning of the data is performed by removing orders that were associated with canceled transactions. Also, orders with negative quantities were removed since such data is not reliable. For each product, revenue is calculated by multiplying the quantity of each item purchased by the price of each item. The data is partitioned by product, and revenue generated by each product is calculated.

Each product is ranked in terms of revenue generated. Also, the top- and bottom-performing products are identified. Additionally, the percentage contribution of the top three products is calculated.

Code

```r
# Data Cleaning
orders_clean <- orders %>%
  filter(!str_detect(InvoiceNo, "^C")) %>%   # remove cancelled orders
  filter(Quantity > 0) %>%                   # remove negative quantities
  mutate(StockCode = toupper(StockCode))    # standardize stock codes

products <- products %>%
  mutate(StockCode = toupper(StockCode))

# Join datasets
data <- orders_clean %>%
  left_join(products, by = "StockCode")

# Remove unwanted descriptions
data <- data %>%
  filter(!Description %in% c("Adjust bad debt", "adjustment", "thrown away")) %>%
  filter(!is.na(Description))

# Calculate revenue
data <- data %>%
  mutate(Revenue = Quantity * UnitPrice)

# Revenue by product
product_revenue <- data %>%
  group_by(StockCode, Description) %>%
  summarise(TotalRevenue = sum(Revenue), .groups = "drop") %>%
  arrange(desc(TotalRevenue))

# Top 10 products
top10 <- product_revenue %>%
  slice_max(TotalRevenue, n = 10)
```

```r
# Bottom 10 products
bottom10 <- product_revenue %>%
  slice_min(TotalRevenue, n = 10)

# Top 3 contribution
top3 <- product_revenue %>%
  slice_max(TotalRevenue, n = 3)

total_revenue <- sum(product_revenue$TotalRevenue)

top3_percentage <- sum(top3$TotalRevenue) / total_revenue * 100

# Display tables
kable(top10, caption = "Top 10 Products by Revenue")
```

Table 1: Top 10 Products by Revenue

StockCode	Description	TotalRevenue
85123A	WHITE HANGING HEART T-LIGHT HOLDER	212942.56
DOT	DOTCOM POSTAGE	206248.77
22423	REGENCY CAKESTAND 3 TIER	174484.74
23843	PAPER CRAFT , LITTLE BIRDIE	168469.60
M	Manual	156225.64
47566	PARTY BUNTING	99504.33
85099B	JUMBO BAG RED RETROSPOT	94340.05
23166	MEDIUM CERAMIC TOP STORAGE JAR	81700.92
POST	POSTAGE	78101.88
23084	RABBIT NIGHT LIGHT	66964.99

```r
kable(bottom10, caption = "Bottom 10 Products by Revenue")
```

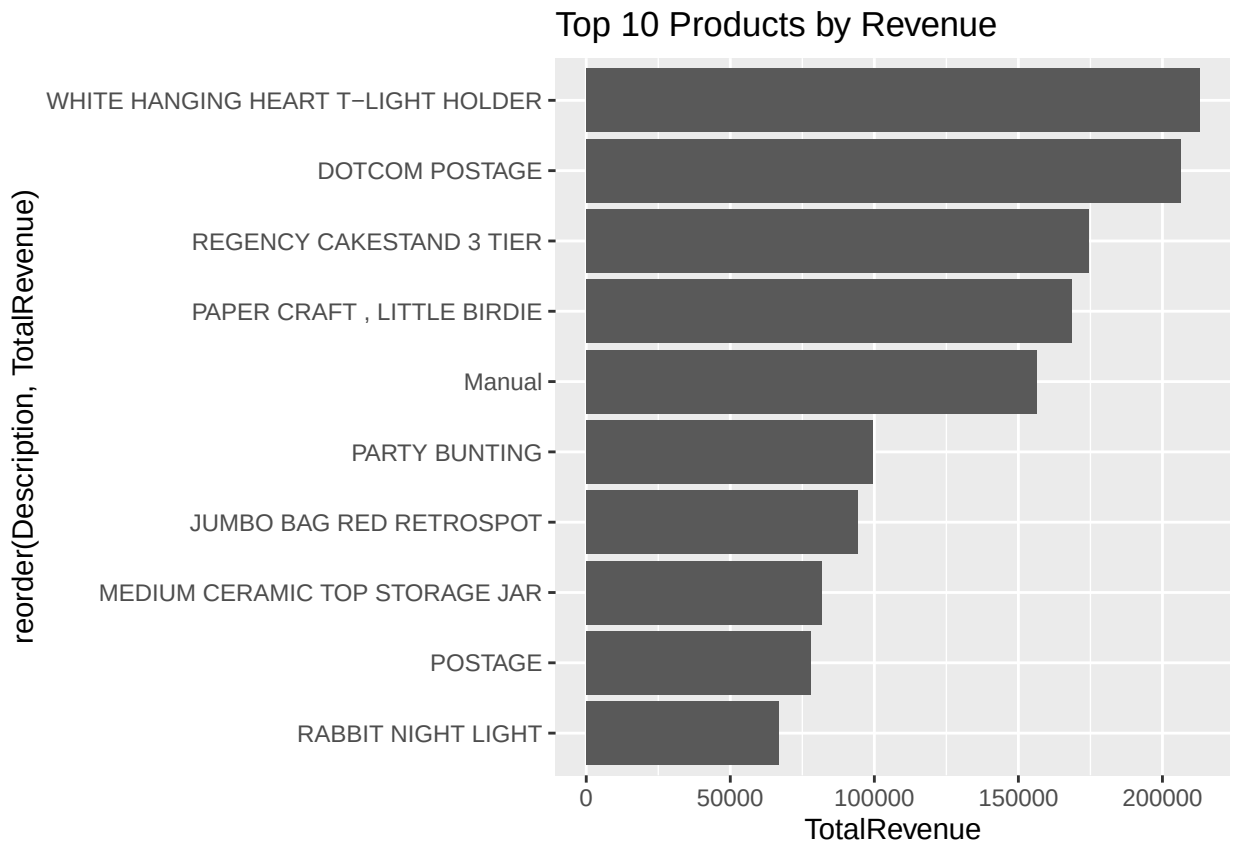
Table 2: Bottom 10 Products by Revenue

StockCode	Description	TotalRevenue
23001	TRAVEL CARD WALLET DOTCOMGIFTSHOP	0.000
35600A	Found by jackie	0.000
PADS	PADS TO MATCH ALL CUSHIONS	0.003
21268	VINTAGE BLUE TINSEL REEL	0.840
90084	PINK CRYSTAL GUITAR PHONE CHARM	0.850
84201C	HAPPY BIRTHDAY CARD TEDDY/CAKE	0.950
84206B	CAT WITH SUNGLASSES BLANK CARD	0.950
21009	ETCHED GLASS STAR TREE DECORATION	1.250
35597A	DUSTY PINK CHRISTMAS TREE 30CM	1.250
35597B	BLACKCHRISTMAS TREE 30CM	1.250
84569C	PACK 4 FLOWER/BUTTERFLY PATCHES	1.250
84743C	ORANGE FELT VASE + FLOWERS	1.250

```r
top3_percentage
```

```
## [1] 5.305872
```

```r
#Visualisation
ggplot(top10, aes(x=reorder(Description,TotalRevenue), y=TotalRevenue))+
  geom_bar(stat="identity")+
  coord_flip()+
  labs(title="Top 10 Products by Revenue")
```



Interpretation

This analysis represent the revenue generated from the sale of a number of products is significant. This indicates that revenue concentration exists, which means that the products that have generated the revenue are the ones that have performed better.

Q2

#Explanation of the Task

The goal of this task is to identify the most valuable customers for the business and analyse their purchasing behaviour.

First, the total spending of each customer is calculated by summing the revenue generated from all of their transactions. Revenue is defined as: Revenue = Quantity * Unit price Cancelled transactions (Invoice numbers starting with "C") must be excluded because they represent returned or cancelled purchases.

After calculating the total spending per customer, the Top 5 highest-spending customers are identified.

Then, for these customers, their purchasing behaviour is analysed by determining which products they purchased the most based on total quantity. This helps the business understand product preferences among its most valuable customers.

#Calculate Customer Spending

```
# Step 1: Remove rows with missing CustomerID
data_clean <- data %>%
  filter(!is.na(CustomerID))

# Step 2: Calculate total spending per customer
customer_spending <- data_clean %>%
  group_by(CustomerID) %>%
  summarise(
    TotalSpending = sum(Revenue, na.rm = TRUE)
  ) %>%
  arrange(desc(TotalSpending))
```

```
class(data)
```

```
## [1] "tbl_df"     "tbl"        "data.frame"
```

```
ls()
```

```
##  [1] "bottom10"          "customer_spending" "data"
##  [4] "data_clean"        "orders"            "orders_clean"
##  [7] "product_revenue"   "products"          "top10"
## [10] "top3"              "top3_percentage"   "total_revenue"
```

```
ls()
```

```
##  [1] "bottom10"          "customer_spending" "data"
##  [4] "data_clean"        "orders"            "orders_clean"
##  [7] "product_revenue"   "products"          "top10"
## [10] "top3"              "top3_percentage"   "total_revenue"
```

#Identify Top 5 Highest Spending Customer

```
top_customers <- customer_spending %>%
  slice_head(n = 5)

# Display result
knitr::kable(top_customers, caption = "Top 5 Highest Spending Customers")
```

Table 3: Top 5 Highest Spending Customers

CustomerID	TotalSpending
14646	287627.1
18102	265212.8
17450	208144.5

CustomerID	TotalSpending
16446	168472.5
14911	153055.5

#Analyse Purchasing Behaviour of Top Customers

```
top_customer_products <- data_clean %>%
  filter(CustomerID %in% top_customers$CustomerID) %>%
  group_by(CustomerID, Description) %>%
  summarise(
    TotalQuantity = sum(Quantity, na.rm = TRUE),
    .groups = "drop"
  )
```

#Identify top products purchased by each customer

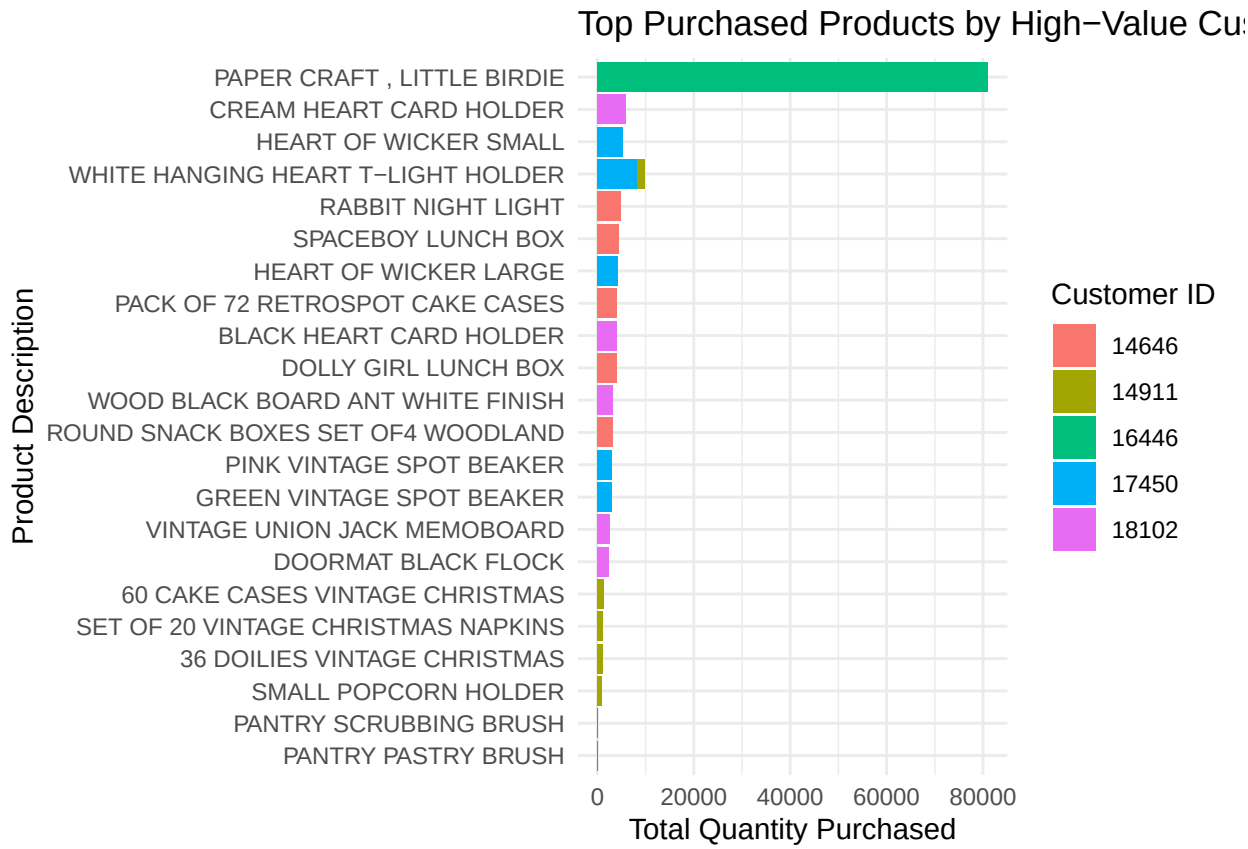
```
top_products_per_customer <- top_customer_products %>%
  group_by(CustomerID) %>%
  slice_max(TotalQuantity, n = 5) %>%
  arrange(CustomerID, desc(TotalQuantity))

# Display table
knitr::kable(top_products_per_customer,
             caption = "Top Purchased Products by High Value Customers")
```

Table 4: Top Purchased Products by High Value Customers

CustomerID	Description	TotalQuantity
14646	RABBIT NIGHT LIGHT	4801
14646	SPACEBOY LUNCH BOX	4492
14646	PACK OF 72 RETROSPOT CAKE CASES	4104
14646	DOLLY GIRL LUNCH BOX	4096
14646	ROUND SNACK BOXES SET OF4 WOODLAND	3120
14911	WHITE HANGING HEART T-LIGHT HOLDER	1540
14911	60 CAKE CASES VINTAGE CHRISTMAS	1440
14911	SET OF 20 VINTAGE CHRISTMAS NAPKINS	1140
14911	36 DOILIES VINTAGE CHRISTMAS	1068
14911	SMALL POPCORN HOLDER	928
16446	PAPER CRAFT , LITTLE BIRDIE	80995
16446	PANTRY PASTRY BRUSH	1
16446	PANTRY SCRUBBING BRUSH	1
17450	WHITE HANGING HEART T-LIGHT HOLDER	8228
17450	HEART OF WICKER SMALL	5286
17450	HEART OF WICKER LARGE	4282
17450	GREEN VINTAGE SPOT BEAKER	3056
17450	PINK VINTAGE SPOT BEAKER	3056
18102	CREAM HEART CARD HOLDER	5946
18102	BLACK HEART CARD HOLDER	4104
18102	WOOD BLACK BOARD ANT WHITE FINISH	3205
18102	VINTAGE UNION JACK MEMOBOARD	2600

CustomerID	Description	TotalQuantity
18102	DOORMAT BLACK FLOCK	2426

```r
#Visualization

top_customer_products <- data_clean %>%
  filter(CustomerID %in% top_customers$CustomerID) %>%
  group_by(CustomerID, Description) %>%
  summarise(TotalQuantity = sum(Quantity), .groups="drop")

top_products_per_customer <- top_customer_products %>%
  group_by(CustomerID) %>%
  slice_max(TotalQuantity, n = 5)
library(ggplot2)

ggplot(top_products_per_customer,
       aes(x = reorder(Description, TotalQuantity),
           y = TotalQuantity,
           fill = factor(CustomerID))) +
  geom_bar(stat = "identity") +
  coord_flip() +
  labs(
    title = "Top Purchased Products by High-Value Customers",
    x = "Product Description",
    y = "Total Quantity Purchased",
    fill = "Customer ID"
  ) +
  theme_minimal()
```




```r
# Check dataset structure
str(data_clean)
```

```
## tibble [412,426 x 9] (S3: tbl_df/tbl/data.frame)
##  $ InvoiceNo  : chr [1:412426] "536365" "536365" "536365" "536365" ...
##  $ StockCode  : chr [1:412426] "85123A" "85123A" "71053" "84406B" ...
##  $ Quantity   : num [1:412426] 6 6 6 8 6 6 2 6 6 6 ...
##  $ InvoiceDate: chr [1:412426] "1/12/2010 8:26" "1/12/2010 8:26" "1/12/2010 8:26" "1/12/2010 8:26" .
##  $ UnitPrice  : num [1:412426] 2.55 2.55 3.39 2.75 3.39 3.39 7.65 4.25 1.85 1.85 ...
##  $ CustomerID : num [1:412426] 17850 17850 17850 17850 17850 ...
##  $ Country    : chr [1:412426] "United Kingdom" "United Kingdom" "United Kingdom" "United Kingdom" .
##  $ Description: chr [1:412426] "WHITE HANGING HEART T-LIGHT HOLDER" "WHITE HANGING HEART T-LIGHT HOL
##  $ Revenue    : num [1:412426] 15.3 15.3 20.3 22 20.3 ...
```

```r
# Check number of unique customers
n_distinct(data_clean$CustomerID)
```

```
## [1] 4339
```

```r
# Verify top customers
print(top_customers)
```

```
## # A tibble: 5 x 2
##   CustomerID TotalSpending
```

```
##        <dbl>         <dbl>
## 1      14646       287627.
## 2      18102       265213.
## 3      17450       208144.
## 4      16446       168472.
## 5      14911       153056.
```

#Interpretation

The results show that a small number of customers generate a significant portion of the company's revenue. The top five customers have spent considerably more than the average customer, indicating that they are highly valuable to the business.

By analysing their purchasing behaviour, it becomes clear that these customers frequently purchase specific products in large quantities. This suggests strong product preferences among high-value customers.

From a business perspective, these customers are critical for maintaining revenue growth. The company could benefit from implementing loyalty programs, personalised promotions, or exclusive offers to retain these high-value customers. Additionally, ensuring that the products frequently purchased by these customers remain available in stock can help maintain customer satisfaction and continued spending.

#Q3 #(a) Explanation

The aim of this task is to analyse product consumption across different countries to understand global market demand.

First, the total revenue generated by each country is calculated. Revenue is defined as: Revenue = Quantity * Unit price The dataset already contains the calculated revenue variable from the previous task. Cancelled transactions were removed during the data cleaning stage, so only valid transactions are included in the analysis.

The data is grouped by Country to determine the total revenue generated in each country. This allows us to identify which countries contribute the most to overall sales.

Next, product consumption is analysed by identifying the single product with the highest total quantity sold in each country. This helps determine which products are most popular in different regions.

Finally, the results are presented in a summary table and visualised using a bar chart to compare revenue across countries.

```r
# Calculate total revenue by country
country_revenue <- data %>%
  group_by(Country) %>%
  summarise(TotalRevenue = sum(Revenue, na.rm = TRUE)) %>%
  arrange(desc(TotalRevenue))

# Display table
kable(country_revenue, caption = "Total Revenue by Country")
```

Table 5: Total Revenue by Country

Country	TotalRevenue
United Kingdom	9440957.77
EIRE	298532.43
Netherlands	293140.27
Germany	239772.03
France	225434.63

Country	TotalRevenue
Australia	141021.41
Spain	72565.62
Switzerland	59581.72
Belgium	42169.87
Norway	39265.02
Sweden	38596.50
Portugal	38545.59
Japan	37905.97
Singapore	33651.53
Finland	28471.75
Hong Kong	22515.58
Channel Islands	21434.89
Denmark	19862.15
Italy	18153.16
Cyprus	14791.93
Austria	11109.38
Israel	8168.37
Poland	7642.00
Iceland	4817.30
Unspecified	4800.55
Greece	4795.62
USA	3778.84
Canada	3695.10
Malta	2926.49
United Arab Emirates	1967.68
Lithuania	1780.46
Lebanon	1711.58
European Community	1459.65
Brazil	1197.15
RSA	1021.78
Czech Republic	826.74
Bahrain	824.94
Saudi Arabia	145.92

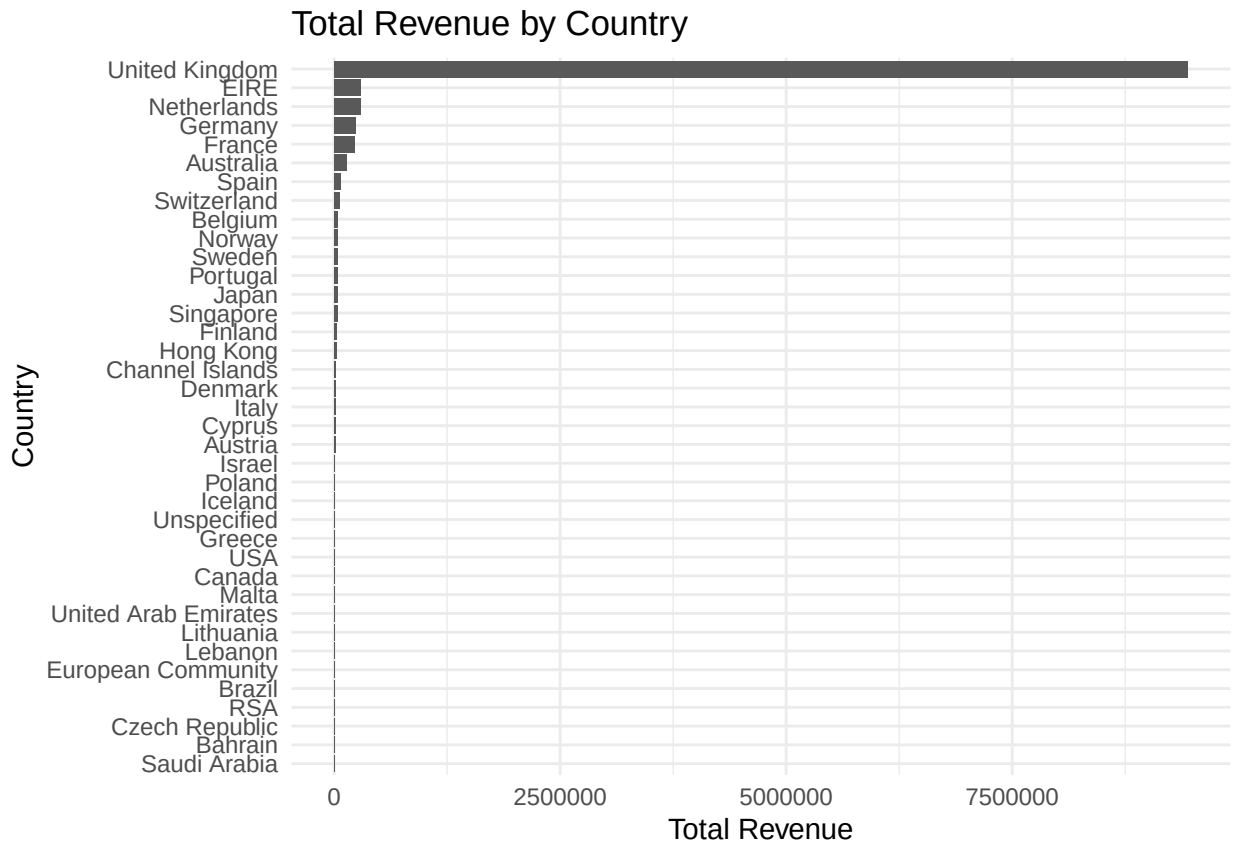
```r
# Calculate quantity sold for each product in each country
country_products <- data %>%
  group_by(Country, Description) %>%
  summarise(TotalQuantity = sum(Quantity, na.rm = TRUE), .groups = "drop")

# Identify most sold product in each country
top_product_country <- country_products %>%
  group_by(Country) %>%
  slice_max(TotalQuantity, n = 1)

# Display table
kable(top_product_country, caption = "Most Popular Product by Country")
```

Table 6: Most Popular Product by Country

Country	Description	TotalQuantity
Australia	MINI PAINT SET VINTAGE	2952
Austria	SET 12 KIDS COLOUR CHALK STICKS	288
Bahrain	ICE CREAM SUNDAE LIP GLOSS	96
Belgium	FAIRY CAKES NOTEBOOK A7 SIZE	640
Brazil	DOLLY GIRL LUNCH BOX	24
Brazil	GREEN REGENCY TEACUP AND SAUCER	24
Brazil	PINK REGENCY TEACUP AND SAUCER	24
Brazil	ROSES REGENCY TEACUP AND SAUCER	24
Brazil	SET OF 4 PANTRY JELLY MOULDS	24
Brazil	SET OF 6 SPICE TINS PANTRY DESIGN	24
Brazil	SET/3 RED GINGHAM ROSE STORAGE BOX	24
Brazil	SMALL HEART FLOWERS HOOK	24
Canada	RETRO COFFEE MUGS ASSORTED	504
Channel Islands	RAIN PONCHO	407
Cyprus	HEART DECORATION PAINTED ZINC	384
Czech Republic	WOODEN STAR CHRISTMAS SCANDINAVIAN	72
Czech Republic	WOODEN TREE CHRISTMAS SCANDINAVIAN	72
Denmark	RED HARMONICA IN BOX	288
EIRE	WHITE HANGING HEART T-LIGHT HOLDER	2184
European Community	CHARLIE AND LOLA FIGURES TINS	24
European Community	CHOCOLATE 1 WICK MORRIS BOX CANDLE	24
European Community	RED ROCKING HORSE HAND PAINTED	24
European Community	ROCKING HORSE GREEN CHRISTMAS	24
European Community	ROCKING HORSE RED CHRISTMAS	24
European Community	SET OF 60 PANTRY DESIGN CAKE CASES	24
European Community	SET OF 60 VINTAGE LEAF CAKE CASES	24
European Community	SET/3 VANILLA SCENTED CANDLE IN BOX	24
European Community	WHITE ROCKING HORSE HAND PAINTED	24
Finland	PINK 3 PIECE POLKADOT CUTLERY SET	1104
France	RABBIT NIGHT LIGHT	4024
Germany	ROUND SNACK BOXES SET OF4 WOODLAND	1233
Greece	4 LAVENDER BOTANICAL DINNER CANDLES	48
Greece	4 PEAR BOTANICAL DINNER CANDLES	48
Hong Kong	PINK 3 PIECE POLKADOT CUTLERY SET	240
Iceland	3D DOG PICTURE PLAYING CARDS	252
Israel	WOODLAND CHARLOTTE BAG	130
Italy	FEATHER PEN,COAL BLACK	336
Japan	RABBIT NIGHT LIGHT	3408
Lebanon	ASSTD FRUIT+FLOWERS FRIDGE MAGNETS	24
Lithuania	FELTCRAFT DOLL ROSIE	48
Lithuania	IVORY ENCHANTED FOREST PLACEMAT	48
Lithuania	RED HARMONICA IN BOX	48
Malta	GRAND CHOCOLATECANDLE	81
Netherlands	RABBIT NIGHT LIGHT	4801
Norway	SMALL FOLDING SCISSOR(POINTED EDGE)	576
Poland	STRAWBERRY CERAMIC TRINKET BOX	168
Portugal	mailout	376
RSA	mailout	20
Saudi Arabia	ASSORTED BOTTLE TOP MAGNETS	12
Saudi Arabia	HOMEMADE JAM SCENTED CANDLES	12

Country	Description	TotalQuantity
Saudi Arabia	PLASTERS IN TIN CIRCUS PARADE	12
Saudi Arabia	PLASTERS IN TIN SKULLS	12
Saudi Arabia	PLASTERS IN TIN STRONGMAN	12
Singapore	CHRISTMAS TREE PAINTED ZINC	384
Spain	PINK 3 PIECE POLKADOT CUTLERY SET	2178
Sweden	MINI PAINT SET VINTAGE	2916
Switzerland	PLASTERS IN TIN WOODLAND ANIMALS	639
USA	SET 12 COLOURING PENCILS DOILY	88
United Arab Emirates	ASSORTED CHEESE FRIDGE MAGNETS	72
United Arab Emirates	BIG DOUGHNUT FRIDGE MAGNETS	72
United Kingdom	PAPER CRAFT , LITTLE BIRDIE	80995
Unspecified	WORLD WAR 2 GLIDERS ASSTD DESIGNS	96

```r
#Revenue Comparison Across Countries
ggplot(country_revenue,
       aes(x = reorder(Country, TotalRevenue),
           y = TotalRevenue)) +
  geom_bar(stat = "identity") +
  coord_flip() +
  labs(
    title = "Total Revenue by Country",
    x = "Country",
    y = "Total Revenue"
  ) +
  theme_minimal()
```



#Interpretation

The results show clear differences in revenue generation across countries. The United Kingdom generates the highest revenue, which is expected because the retailer is based in the UK and likely has a larger customer base there.

Other European countries such as Germany, France, and the Netherlands also contribute significantly to the company's revenue. This indicates that the retailer has a strong presence in the European market.

The analysis of product consumption shows that different countries have different product preferences. Some products are extremely popular in specific countries, suggesting that cultural preferences or seasonal demand may influence purchasing behaviour.

From a business perspective, these insights can help the company:

Identify high-performing international markets

Adjust marketing strategies for different countries

Promote products that are popular in specific regions

Expand operations in countries with strong demand

Understanding regional purchasing behaviour allows the business to better target customers and improve global sales performance.

#Q4 #(a) Explanation

The purpose of this task is to analyse sales trends over time using the transaction date variable InvoiceDate.

First, the month and year are extracted from the InvoiceDate variable to allow analysis of sales over time. This allows transactions to be grouped by month and year.

Next, the total revenue for each month is calculated by summing the revenue generated from all transactions in that month. This helps identify which months generate the highest sales.

In addition, the analysis identifies products that consistently appear among the lowest-selling products across multiple months. This helps determine which products perform poorly over time and may require review by the business.

Finally, the monthly revenue trend is visualised using a time-series line plot to observe any patterns or seasonal effects in sales.

```r
# Load libraries
library(dplyr)
library(lubridate)
library(ggplot2)
library(knitr)

# Convert InvoiceDate to datetime and remove invalid dates
data <- data %>%
  mutate(InvoiceDate = dmy_hm(InvoiceDate)) %>%
  filter(!is.na(InvoiceDate))

# Create Year, Month, and YearMonth variables
data <- data %>%
  mutate(
    Year = year(InvoiceDate),
    Month = month(InvoiceDate),
    YearMonth = floor_date(InvoiceDate, "month")
  )

# Inspect
head(data)
```

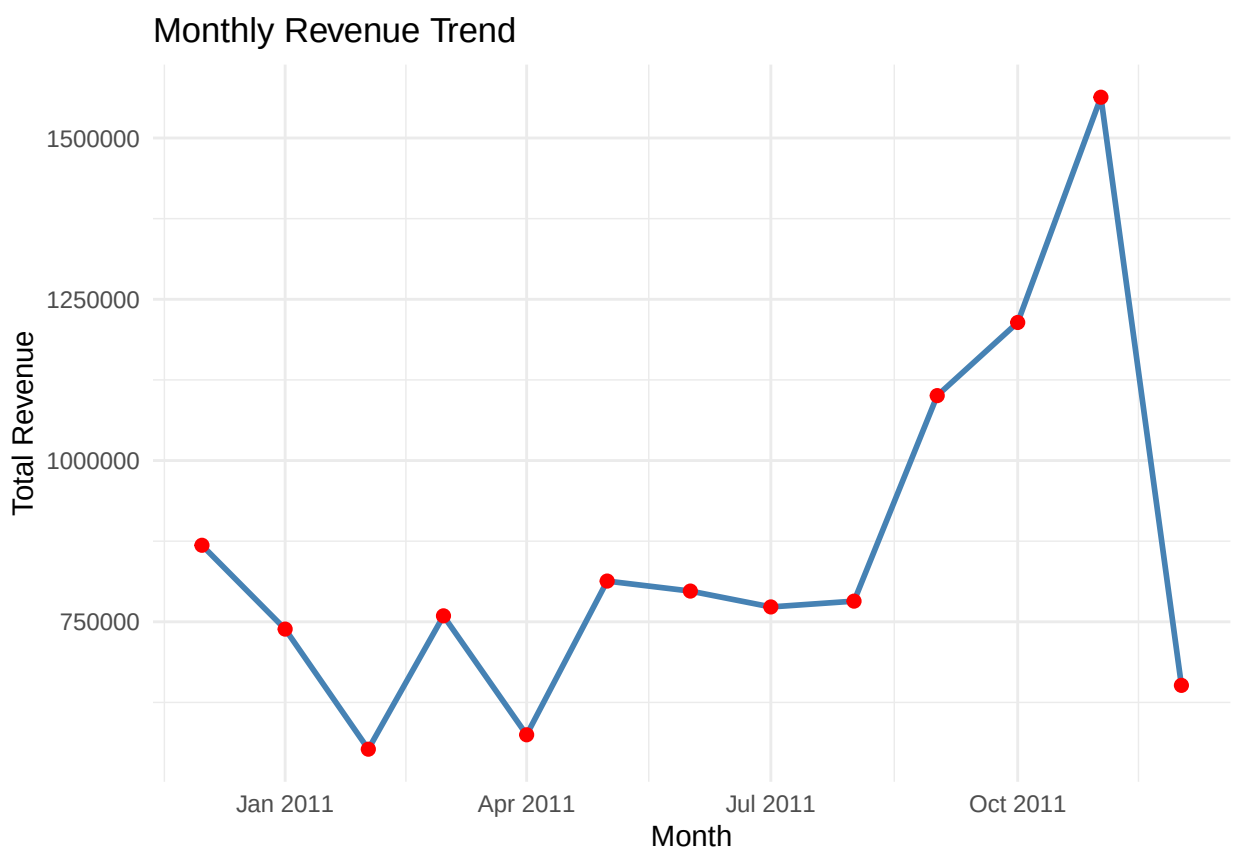
```
## # A tibble: 6 x 12
##   InvoiceNo StockCode Quantity InvoiceDate         UnitPrice CustomerID Country
##   <chr>     <chr>        <dbl> <dttm>                  <dbl>      <dbl> <chr>
## 1 536365    85123A           6 2010-12-01 08:26:00      2.55      17850 United ~
## 2 536365    85123A           6 2010-12-01 08:26:00      2.55      17850 United ~
## 3 536365    71053            6 2010-12-01 08:26:00      3.39      17850 United ~
## 4 536365    84406B           8 2010-12-01 08:26:00      2.75      17850 United ~
## 5 536365    84029G           6 2010-12-01 08:26:00      3.39      17850 United ~
## 6 536365    84029E           6 2010-12-01 08:26:00      3.39      17850 United ~
## # i 5 more variables: Description <chr>, Revenue <dbl>, Year <dbl>,
## #   Month <dbl>, YearMonth <dttm>
```

```r
colnames(data)
```

```
##  [1] "InvoiceNo"   "StockCode"   "Quantity"    "InvoiceDate" "UnitPrice"
##  [6] "CustomerID"  "Country"     "Description" "Revenue"     "Year"
## [11] "Month"       "YearMonth"
```

```r
# Calculate monthly revenue
monthly_revenue <- data %>%
  group_by(YearMonth) %>%
  summarise(TotalRevenue = sum(Revenue, na.rm = TRUE),
```

```r
            .groups = "drop")

# Plot monthly revenue trend
ggplot(monthly_revenue, aes(x = YearMonth, y = TotalRevenue)) +
  geom_line(color = "steelblue", size = 1) +
  geom_point(color = "red", size = 2) +
  labs(
    title = "Monthly Revenue Trend",
    x = "Month",
    y = "Total Revenue"
  ) +
  theme_minimal()
```




```r
# Check range and summary
range(monthly_revenue$YearMonth)
```

```
## [1] "2010-12-01 UTC" "2011-12-01 UTC"
```

```r
summary(monthly_revenue$YearMonth)
```

```
##                  Min.               1st Qu.                Median
## "2010-12-01 00:00:00" "2011-03-01 00:00:00" "2011-06-01 00:00:00"
##                  Mean               3rd Qu.                  Max.
## "2011-06-01 07:23:04" "2011-09-01 00:00:00" "2011-12-01 00:00:00"
```

```r
#Identify Highest Sales Months
top_months <- monthly_revenue %>%
  arrange(desc(TotalRevenue)) %>%
  slice_head(n = 5)

kable(top_months, caption = "Months with Highest Sales")
```

Table 7: Months with Highest Sales

YearMonth	TotalRevenue
2011-11-01	1563226
2011-10-01	1214040
2011-09-01	1100625
2010-12-01	868633
2011-05-01	813138

```r
# Identify top 5 highest revenue months
top_months <- monthly_revenue %>%
  arrange(desc(TotalRevenue)) %>%
  slice_head(n = 5)

# Display
kable(top_months, caption = "Months with Highest Sales")
```

Table 8: Months with Highest Sales

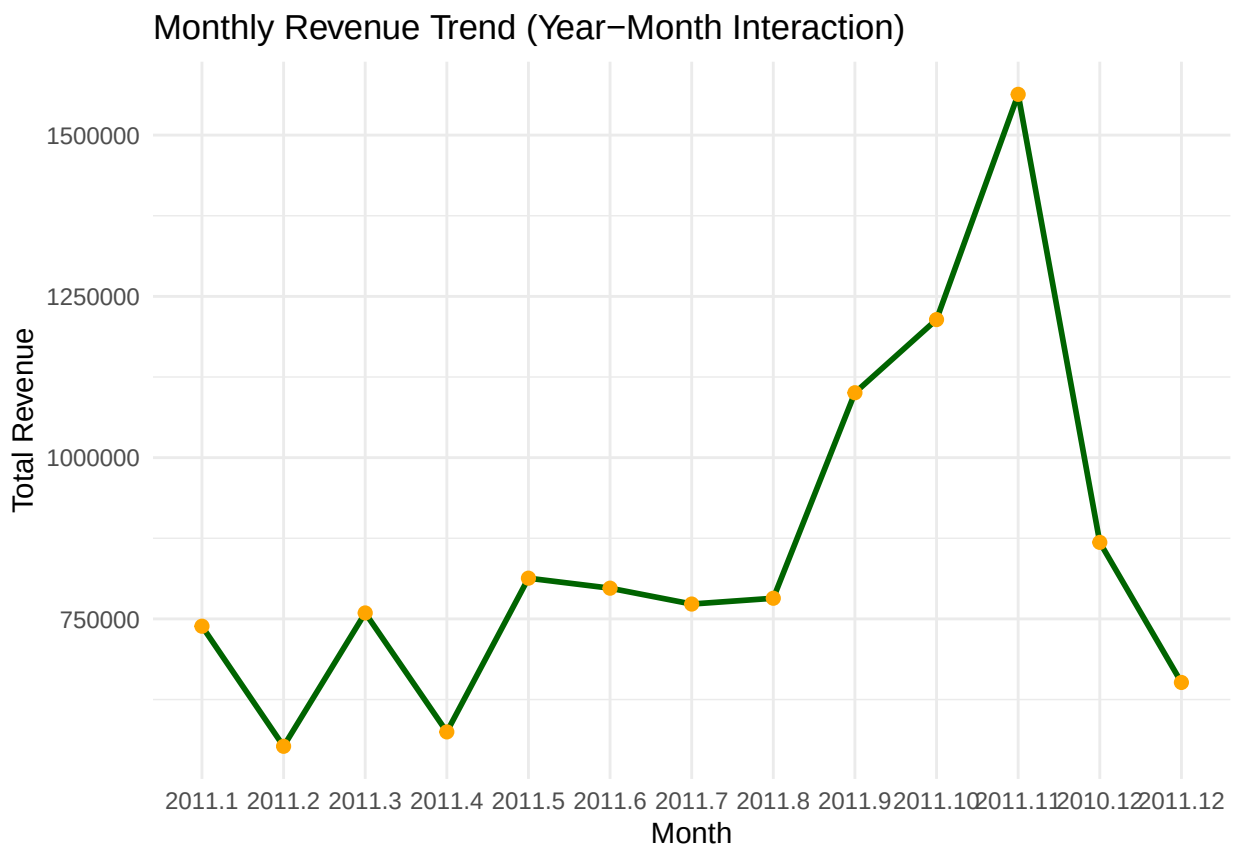
YearMonth	TotalRevenue
2011-11-01	1563226
2011-10-01	1214040
2011-09-01	1100625
2010-12-01	868633
2011-05-01	813138

```r
# Calculate monthly revenue per product
monthly_product_sales <- data %>%
  group_by(Year, Month, Description) %>%
  summarise(
    MonthlyRevenue = sum(Revenue, na.rm = TRUE),
    .groups = "drop"
  )

# Identify lowest-selling product per month
lowest_products <- monthly_product_sales %>%
  group_by(Year, Month) %>%
  slice_min(MonthlyRevenue, n = 1, with_ties = FALSE)

# Display
kable(lowest_products, caption = "Lowest Selling Products per Month")
```

Table 9: Lowest Selling Products per Month

Year	Month	Description	MonthlyRevenue
2010	12	LUNCH BOX WITH CUTLERY FAIRY CAKES	0
2011	1	OLD DOC RUSSEL METAL SIGN	0
2011	2	DOORMAT MERRY CHRISTMAS RED	0
2011	3	DIAMANTE BOW BROOCH RED COLOUR	0
2011	4	ACRYLIC HANGING JEWEL,BLUE	0
2011	5	ANTIQUE GLASS HEART DECORATION	0
2011	6	ANTIQUE GLASS PLACE SETTING	0
2011	7	?	0
2011	8	STRAWBERRY DREAM CHILDS UMBRELLA	0
2011	9	CREAM AND PINK FLOWERS PONY	0
2011	10	ACRYLIC JEWEL SNOWFLAKE, PINK	0
2011	11	BLUE GIANT GARDEN THERMOMETER	0
2011	12	BLUE GIANT GARDEN THERMOMETER	0

```r
# Optional: Monthly revenue trend with Year-Month interaction
monthly_revenue_interaction <- data %>%
  group_by(Year, Month) %>%
  summarise(TotalRevenue = sum(Revenue, na.rm = TRUE),
            .groups = "drop")

ggplot(monthly_revenue_interaction,
       aes(x = interaction(Year, Month),
           y = TotalRevenue,
           group = 1)) +
  geom_line(color = "darkgreen", size = 1) +
  geom_point(color = "orange", size = 2) +
  labs(
    title = "Monthly Revenue Trend (Year-Month Interaction)",
    x = "Month",
    y = "Total Revenue"
  ) +
  theme_minimal()
```



#Interpretation

The monthly revenue analysis reveals fluctuations in sales across different months. Some months show significantly higher revenue compared to others, suggesting the presence of seasonal demand patterns.

For example, higher sales may occur during holiday seasons such as November and December, when customers tend to purchase more gifts and seasonal products. Lower sales months may occur during periods with less consumer spending.

The analysis of lowest-selling products across months indicates that some products consistently perform poorly. These products may have limited demand, outdated designs, or may not appeal to the majority of customers.

From a business perspective, these insights can help the company:

Identify seasonal sales trends

Improve inventory planning

Plan promotional campaigns during low-sales periods

Consider discontinuing or redesigning consistently low-performing products

Understanding sales trends over time allows the business to make better strategic decisions and optimise revenue throughout the year.